

Wavelength Selection Considerations for Optical Wireless Positioning Systems

Jorik De Bruycker
KU Leuven, WaveCore, ESAT
Ghent Technology Campus
Ghent, Belgium
jorik.debruycker@kuleuven.be

Frédéric B. Leloup
KU Leuven, WaveCore, ESAT
Ghent Technology Campus
Ghent, Belgium
frederic.leloup@kuleuven.be

Nobby Stevens
KU Leuven, WaveCore, ESAT
Ghent Technology Campus
Ghent, Belgium
nobby.stevens@kuleuven.be

Abstract—Indoor Positioning Systems act as an important technology to provide real-time location estimation, enabling a large variety of industrial applications. In this context, Optical Wireless Positioning employs the propagation characteristics of optical signals as a means to calculate an accurate and precise position. While Visible Light Positioning focuses on the use of LED lighting to support both positioning and illumination simultaneously, Infrared-based systems also offer viable positioning solutions featuring distinct advantages and drawbacks over their visible light counterparts. The selection of the used wavelength thus proves to be an important design consideration for an Optical Wireless Positioning system. This work summarises the main differences and trade-offs fundamental to the selection of the wavelength, and compares them in order to make an informed decision on the selection.

I. INTRODUCTION

Optical Wireless Positioning (OWP) is a technology within the larger domain of Optical Wireless Communications (OWC). OWP systems are exploiting the distinctive properties of optical signals (compared to RF signals [1]) to provide accurate and precise positioning for indoor environments [2]. In the case where the visible light spectrum is employed, the term Visible Light Positioning (VLP) is used [3]–[5]. However, Infrared (IR) and more specifically Near-Infrared (NIR) are also often utilized [6], [7]. The selection of the wavelength for OWP systems is of high importance and inherently comes with a number of trade-offs that should be considered when designing and deploying such a system.

Remark that the majority of the different advantages and drawbacks presented in this work are also applicable to OWC in general. However, depending on the intended solution, other design requirements are demanded. For bidirectional optical wireless communications for instance [8], visible light is used in the downlink channel to provide the illumination functionality simultaneously with communication, while infrared is applied in the uplink in order to avoid blinding the users. Another technology belonging to OWC is free space optics (FSO) [9]. Here, the impact of possible adverse weather conditions forms yet another important selection criterion, as the channel is outdoors. It has been demonstrated both theoretically and experimentally, that the impact of fog and

rain is less detrimental for infrared light compared to visible light as carrier of the information [10], [11].

This work is meant to provide an overview of the specific considerations for OWP and compares the advantages and disadvantages inherent to visible and IR wavelengths in this scenario. The distinguishing parameters that are significant in the selection of the system operating wavelength are discussed in Section II, while Section III summarizes the main takeaways of this work and provides a conclusion.

II. SELECTION CRITERIA FOR VISIBLE OR INFRARED LEDs

A. Functionality and Infrastructure Requirements

The progress in LED technology features major advantages over traditional lighting solutions including longer lifespan, decreased cost and higher efficiency [12]–[14]. These beneficial factors have lead to its rapid and widespread adoption, covering about half of the deployed lighting in buildings in 2020 and projected to fully dominate the lighting market by 2025 [15]. On top of these distinct advantages compared to more traditional lighting elements, LEDs also have a higher optical bandwidth. As this bandwidth is well beyond the flicker threshold of the human eye [16], the possibility to impose a dedicated intensity modulation scheme on top of the illumination functionality enables VLP. This development is a significant driver enabling the deployment of VLP systems. By employing LEDs that act both as lighting infrastructure and as beacons for indoor positioning, a dual functionality is achieved [17], [18]. This aspect is a major asset for VLP systems, yet it also brings forth a number of challenges and limitations as will be discussed in the following sections. While IR-based systems do not suffer from these limitations to the same extent, they do however require dedicated beacons in addition to the already required indoor lighting infrastructure, increasing costs. Using visible light LEDs on the other hand also means that the illumination should always be powered on in order to provide the positioning functionality, which might not always be desirable or necessary.

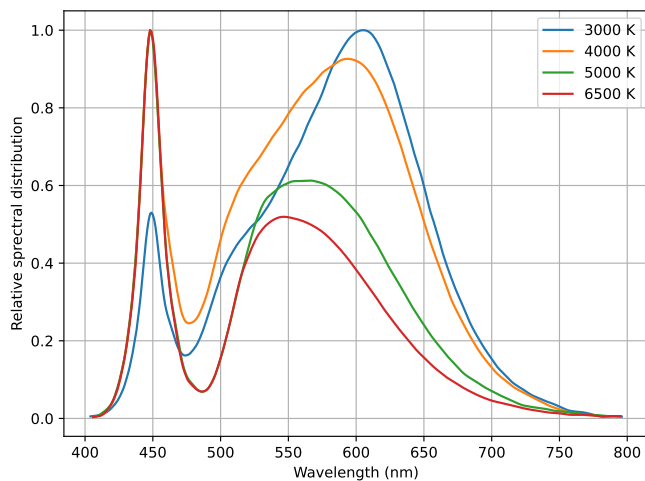


Fig. 1: Normalized spectral distribution for Bridgelux V15 series white LEDs with different color temperatures [20].

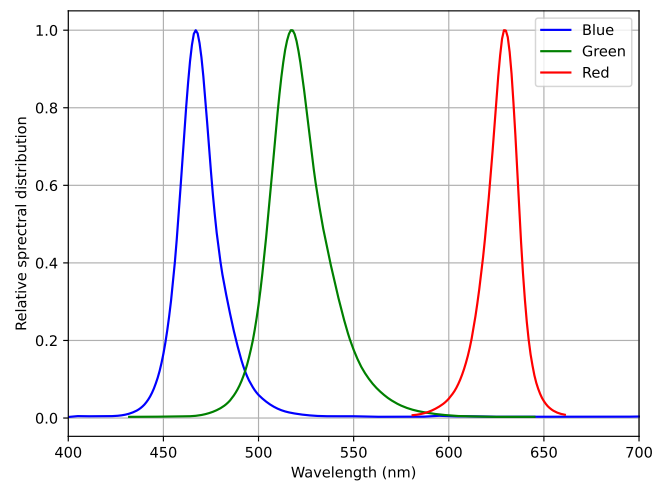


Fig. 2: Normalized spectral distribution for different RGB channels of Luxeon 3528 RGB series LEDs [21].

B. Operating Voltage

The wavelength of the light emitted by an LED is determined by the bandgap of the semiconductor material [19]. Shorter wavelengths correspond to a higher energy and bandgap and as a consequence lead to increased operation voltages. As IR corresponds to higher wavelengths, the corresponding LEDs thus typically have a lower operating voltage in comparison to visible light alternatives at the same current. As LEDs are often arranged in a combination of series and parallel strings, the operating voltage can be controlled to a certain extent. Nevertheless, this difference has to be considered when designing or selecting an appropriate LED driver while this implies that the visible light and IR LEDs are not necessarily interchangeable for the same LED driver.

C. Bandwidth Considerations

1) *Visible Light LEDs:* When solely focusing on the time variation of the visible light intensity, it can be said that modulating visible light implies a crucial health related constraint on the minimal perceivable frequency. Studies show that frequencies above 200Hz cause no physical response and thus may not cause any harm [16]. The IEEE Standard 802.15.7 on Visible Light Communication therefore defines the maximum flicker time period (MFTP), a time frame in which the average light intensity should remain constant, as 5 ms [23]. This constraint puts a lower limit on the modulation frequency for visible light and has the consequence that special care needs to be taken for the modulation format to assure a constant human perceived illumination over time.

The upper limit in modulation frequency is limited by the selected transmitter. In general, the vast majority of white LEDs are fabricated using blue LEDs with a yellow phosphor coating [24]. This is very cost effective and enhances the

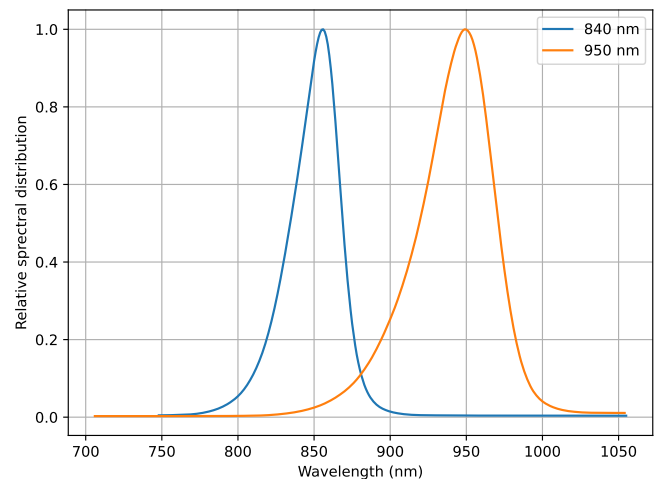


Fig. 3: Normalized spectral distribution for Luxeon IR Domed Line LEDs [22].

color stability over time, yet the response time of the phosphorescent component is slower compared to the blue LED's contribution. In practice this typically limits the maximum optical bandwidth to several MHz for traditional white LEDs [25]–[27]. Fig. 1 illustrates the normalized spectral distribution for a commercial white phosphor LED available with different thicknesses of phosphor layers targeting several color temperatures. Lower color temperatures correspond to higher relative power in the yellow wavelengths due to phosphorence compared to the blue wavelength range. A blue filter at the receiver can mitigate the detrimental effect of the phosphor coating on the bandwidth [28], yet it also reduces the received intensity. Another solution mixes the intensities of RGB-LEDs to be perceived as light of a desired color. The normalized

spectral distribution for the different channels of a commercial RGB LED is shown in Fig. 2. While the lack of the phosphor layer entails a higher bandwidth [29], [30], this solution complicates the LED driver requirements and increases costs.

2) *Infrared LEDs*: Obviously, due to the fact that IR can not be perceived by the human eye, the constraints on perceivable flicker do not apply to infrared systems, thus allowing a much higher degree of freedom in the modulation scheme as the need to respect the MFTP does not apply. Infrared LEDs generally achieve higher bandwidths [31], but do not come with the additional cost and complexity as with RGB-LEDs.

For NIR, wavelengths of 850 nm and 940 nm are often used [32]. The normalized spectral distributions for two commercial IR LEDs operating at these wavelengths are presented in Fig. 3. While the emitted spectrum is still broad, 850 nm emitters can have a red glare due to their small yet sufficient to be human observable contribution in the red wavelength range. Depending on the application requirements, this can either prove to be convenient in order to verify whether the LEDs are active and correctly working as inactive or malfunctioning LEDs can be visibly observed by the human eye. Yet, it can also be undesirable to perceive any red glare. LEDs emitting at 940 nm are however fully invisible to the human eye and may be favorable in this regard.

D. Regulations and Eye Safety

With respect to health and safety, exposure limit values for artificial non-coherent optical radiation to human eyes and skin have been laid down in an European Directive (2006/25/EC) [33]. The International Standard IEC 62471 gives guidance for evaluating the photobiological safety of lamps and lamp systems, including a standardized technique for evaluating potential radiation hazards [34]. One important hazard associated to visible light, in particular to the blue light contribution (400 to 490 nm), is the so-called blue light hazard, which may lead to photoreinitis or blue light retinal injury. Inappropriate exposure to both visible and IR light might cause retinal burn and skin burn, while exposure to IR light for too long time above a total irradiance threshold may lead to corneal burn. Furthermore, specific lighting requirements for indoor workplaces are defined by an European Standard BS EN 12464-1:2021 [35], including requirements with regard to visual comfort such as Unified Glare Rating (UGR). Respecting these specifications might compromise an optimal functionality of a visible light system and vice versa to a certain extent, which is evidently not the case for IR based systems.

E. Stroboscopic effect

As the deployment of OWP requires modulation of the transmitted optical signals, stroboscopic effects may occur when visible light is employed [36], even when the MFTP defined by IEEE 802.15.7 is respected. In factory settings where machinery rotating at high speeds is present, serious safety hazards may occur as the modulated light may lead

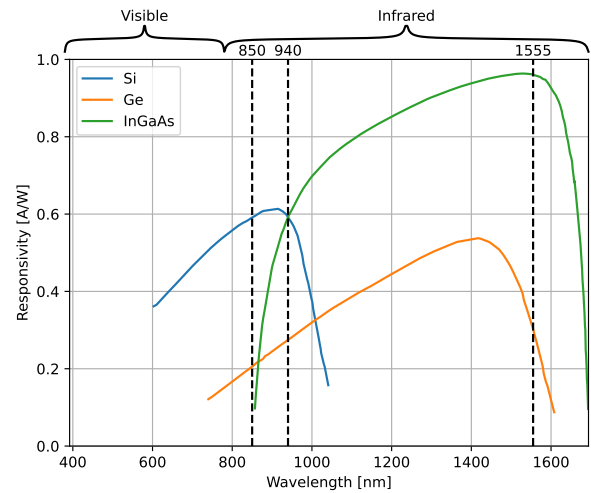


Fig. 4: Typical responsivity of Si, Ge and InGaAs photodetectors [40]. Visible light and near infrared wavelength ranges are indicated by the brackets above, commonly used infrared wavelengths of 850 nm, 940 nm and 1555 nm are marked with the vertical black striped lines.

to workers visually misinterpreting the rotation speed of the machinery, possibly leading to severe accidents. In these conditions, IR LEDs should always be favored over using visible light for safety reasons.

F. Photodiode Responsivity

The responsivity $R_p(\lambda)$ of a photodetector is expressed in A/W, which corresponds to the electrical photocurrent induced by the incident radiant power for a given wavelength. Exemplary responsivities for a silicon, germanium and indium gallium arsenide detector are shown in Fig. 4. It is clear that the selection of the photodetector has to be made in function of the transmitter wavelength. VLP systems generally employ silicon detectors as these have a decent responsivity in the visible light spectrum. However, silicon detectors typically show their peak responsivity in the NIR range, thus favoring the use of infrared transmitters. While blue-enhanced silicon photodiodes can take advantage of the characteristic blue peak in commonplace white phosphor LEDs [37], [38], their peak responsivity can still be found at infrared wavelengths. Furthermore, these white LEDs come in different color temperature variants, associated with a difference in relative intensities for blue and yellow wavelength ranges [39].

G. Temperature Coefficient

The radiant flux emitted by an LED is affected by the LED junction temperature. In case of IR LEDs, this dependency is typically higher compared to their visible light counterparts. For example, Fig. 5 illustrates the temperature dependence for a visible light LED and an IR LED. While both visible light and IR have a non-negligible dependency, thermal performance should be taken into account for both options, yet

even more so for IR systems. Especially for received signal strength based OWP [41], temperature variations can have a detrimental effect.

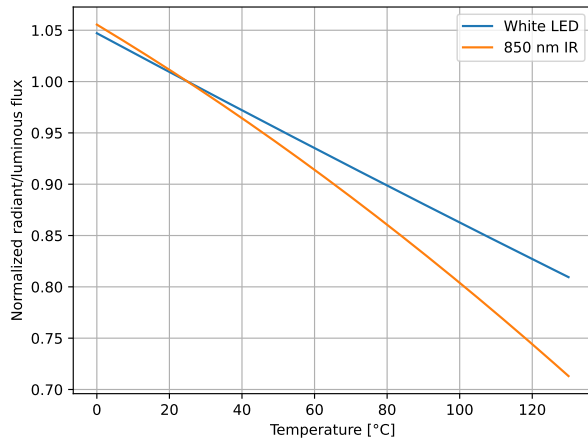


Fig. 5: Comparison between the normalized luminous flux of a Bridgelux V15 Array Series Visible Light LED [20] and the normalized radiant flux of a Luxeon IR Domed Line 850 nm [22] in function of the LED junction temperature. Both curves are normalized with respect to the luminous flux at 25 °C.

H. Color shift

Temperature variations may also lead to a shift in the peak transmitted wavelength [42], typically indicated in the LED datasheet by the increase in nm/°C. On one hand, this might be undesirable for visible light if the color appearance is essential. On the other hand, this should also be taken into account in relation to the photodetector responsivity as described in section II-F. For example, the responsivity for silicon detectors has a positive slope in the visible light spectrum and NIR up until typically 900 - 1000 nm. For wavelengths higher than this range, the responsivity shows a steeper negative slope which implies an increased sensitivity to peak wavelength increase and thus temperature variations.

I. Attenuation due to Dust Accumulation

Over time, OWP receivers will inevitably collect dust, which leads to signal attenuation. In the field of solar panel performance, several works have evaluated the impact of dust in function of the incident wavelength [43], [44], showing a slight difference in signal attenuation favoring IR over visible light. Other works show only a marginal decrease in transmittance with respect to wavelength [45] or describe the change in spectral transmittance to be almost the same [46]. While the differences are small, IR seems to be favorable with regard to robustness to dust accumulation. The main takeaway is however that dust acts as a deteriorating factor for both visible and infrared wavelength based systems. The

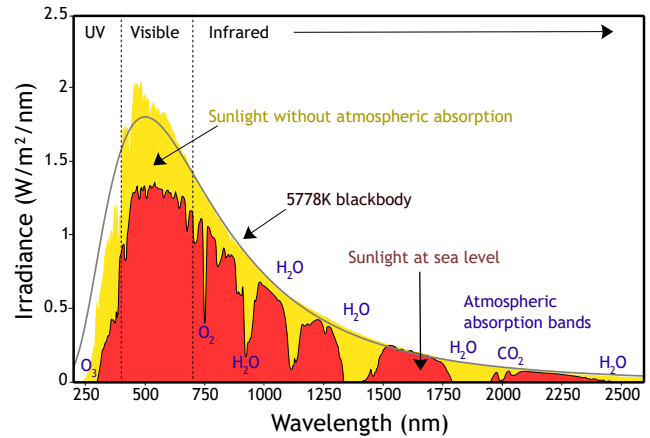


Fig. 6: Solar radiation spectrum without atmospheric absorption (yellow) and at sea level (red) [48]. The visible light wavelength range shows a higher irradiance compared to infrared, while the infrared wavelength also show a drop due to the atmospheric absorption.

comparison for both is a difficult topic that depends heavily on the type and granularity of the dust, which can vary widely across different environments. As the impact can be quite severe, dust deposition on the receiver should be minimized regardless of the transmitter wavelength by avoiding any possible accumulation by regular cleaning and maintenance.

J. Solar Irradiance and Ambient Light

While OWP is meant for indoor positioning, both solar irradiance and ambient light will still inherently be present in realistic scenarios. The solar radiation spectrum both with and without atmospheric absorption is depicted in Fig. 6. It is clear that most irradiance is allocated in the visible light spectrum. For IR wavelengths, the solar irradiance is lower and additionally exhibits several dips due to atmospheric absorption. IR systems are thus preferable with respect to solar irradiance and ambient light, especially when combined with optical infrared bandpass filters. Furthermore, transmission through typical glass windows or skylights in real indoor environments is also subject to a spectral dependence. The majority of glass and films used for windows have a better transmission in the visible light range, while blocking more in the infrared range [47].

K. Summarizing table

In Table I, a summarizing overview can be found, where a “+” denotes favorable, and “-” stands for disadvantageous. Obviously, it concerns a relative comparison between infrared and visible light where these wavelengths are used for an OWP solution. Regarding a visible light deployment, a distinction is made between phosphor coated LEDs (abbreviated as PC in

the table) and RGB LEDs when relevant, this distinction is omitted for selection criteria where it is irrelevant.

TABLE I: Summary of the discussed selection criteria.

	Infrared usage	Visible light usage	
		PC	RGB
Dual functionality	-	+	
Bandwidth	+	-	+
Driver complexity	+	+	-
Eye safety	+	-	
Stroboscopic effect	+	-	
Temperature coefficient	-	+	
Impact dust	+	-	
Ambient light impact	+	-	
Responsivity receiver	+	-	

III. CONCLUSION

This work provides an overview of the main properties that should be considered in the design of an OWP system with regard to the choice for either visible light or infrared transmitters. The direct impact on localization accuracy is difficult to assess as there are a vast number number of characteristic parameters that may contribute to the overall system performance. These parameters are summarized and discussed and should be acknowledged when designing novel OWP systems. The main appealing feature of systems employing visible light LEDs clearly lies in the dual functionality as they operate as both illumination and positioning beacons simultaneously. However, in certain aspects it is also explained how infrared LEDs can prove to be favorable over visible light LEDs. All summarized parameters should thus be carefully considered and examined against one another with an eye on the specific requirements of the desired positioning system. Finally, it is also worth noting that the discussed parameters are also applicable for OWC systems in general, and should also be considered in the design of aforementioned systems.

REFERENCES

- [1] M. F. Keskin and S. Gezici, "Comparative Theoretical Analysis of Distance Estimation in Visible Light Positioning Systems," *Journal of Lightwave Technology*, vol. 34, no. 3, pp. 854–865, 2016.
- [2] M. Bergen, D. Guerrero, X. Jin, B. Hristovski, H. Chaves, R. Klukas, and J. Holzman, "Design and optimization of indoor optical wireless positioning systems," 03 2016, p. 97540A.
- [3] P. H. Pathak, X. Feng, P. Hu, and P. Mohapatra, "Visible Light Communication, Networking, and Sensing: A Survey, Potential and Challenges," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2047–2077, 2015.
- [4] Y. Zhuang, L. Hua, L. Qi, J. Yang, P. Cao, Y. Cao, Y. Wu, J. Thompson, and H. Haas, "A Survey of Positioning Systems Using Visible LED Lights," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 1963–1988, 2018.
- [5] J. Luo, L. Fan, and H. Li, "Indoor Positioning Systems Based on Visible Light Communication: State of the Art," *IEEE Communications Surveys & Tutorials*, vol. PP, pp. 1–1, 08 2017.
- [6] E. Aitenbichler and M. Mühlhäuser, "An IR Local Positioning System for Smart Items and Devices," 06 2003, pp. 334–339.
- [7] T. Arai, T. Yoshizawa, T. Aoki, K. Zempo, and Y. Okada, "Evaluation of Indoor Positioning System based on Attachable Infrared Beacons in Metal Shelf Environment," in *2019 IEEE International Conference on Consumer Electronics (ICCE)*, 2019, pp. 1–4.
- [8] G. Cossu, R. Corsini, and E. Ciaramella, "High-Speed Bi-directional Optical Wireless System in Non-Directed Line-of-Sight Configuration," *Journal of Lightwave Technology*, vol. 32, no. 10, pp. 2035–2040, 2014.
- [9] Z. Ghassemlooy, S. Arnon, M. Uysal, Z. Xu, and J. Cheng, "Emerging Optical Wireless Communications-Advances and Challenges," *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 9, pp. 1738–1749, 2015.
- [10] M. Ijaz, Z. Ghassemlooy, J. Perez, V. Brazda, and O. Fiser, "Enhancing the Atmospheric Visibility and Fog Attenuation Using a Controlled FSO Channel," *IEEE Photonics Technology Letters*, vol. 25, no. 13, pp. 1262–1265, 2013.
- [11] A. Vavoulas, H. G. Sandalidis, and D. Varoutas, "Weather effects on FSO network connectivity," *Journal of Optical Communications and Networking*, vol. 4, no. 10, pp. 734–740, 2012.
- [12] Z. Ghassemlooy, L. N. Alves, S. Zvánovec, and M.-A. Khalighi, Eds., *Visible Light Communications: Theory and Applications*. CRC Press, Jun. 2017. [Online]. Available: <https://doi.org/10.1201/9781315367330>
- [13] M. Zalesinska and M. Gorczewska, "Comparative study of lighting quality and energy efficiency for various road lighting situations," in *2016 IEEE Lighting Conference of the Visegrad Countries (Lumen V4)*, 2016, pp. 1–5.
- [14] A. Keppens, *Modelling and evaluation of high-power light-emitting diodes for general lighting*, 09 2010.
- [15] IEA, "Lighting – analysis," Nov 2021. [Online]. Available: <https://www.iea.org/reports/lighting>
- [16] S. Berman, D. Greenhouse, I. Bailey, R. Clear, and T. Raasch, "Human electretinogram responses to video displays, fluorescent lighting, and other high frequency sources," *Optometry and vision science : official publication of the American Academy of Optometry*, vol. 68, no. 8, pp. 645–662, August 1991. [Online]. Available: <https://doi.org/10.1097/00006324-199108000-00012>
- [17] M. F. Keskin, A. D. Sezer, and S. Gezici, "Localization via Visible Light Systems," *Proceedings of the IEEE*, vol. 106, no. 6, pp. 1063–1088, 2018.
- [18] T. Kominé and M. Nakagawa, "Fundamental analysis for visible-light communication system using LED lights," *IEEE Transactions on Consumer Electronics*, vol. 50, no. 1, pp. 100–107, 2004.
- [19] G. Held, "Introduction to Light Emitting Diode Technology and Applications," 2008.
- [20] *Bridgelux V15 Array Series*, Bridgelux, 5 2014, rev. 2. [Online]. Available: <https://www.bridgelux.com/products/v-series#specifications>
- [21] *LUXEON 3528 RGB*, Lumileds, 3 2022. [Online]. Available: <https://lumileds.com/products/color-leds/luxeon-3528rgb>
- [22] *LUXEON IR Domed Line*, Lumileds, 7 2019. [Online]. Available: <https://lumileds.com/products/infrared-emitters/luxeon-ir-domed-line/>
- [23] "IEEE Standard for Local and metropolitan area networks—Part 15.7: Short-Range Optical Wireless Communications," *IEEE Std 802.15.7-2018 (Revision of IEEE Std 802.15.7-2011)*, pp. 1–407, 2019.
- [24] C. C. Lin, Y. S. Zheng, H. Y. Chen, C. H. Ruan, G. W. Xiao, and R. S. Liu, "Improving Optical Properties of White LED Fabricated by a Blue LED Chip with Yellow/Red Phosphors," *Journal of The Electrochemical Society*, vol. 157, no. 9, p. H900, 2010. [Online]. Available: <https://doi.org/10.1149/1.3465654>
- [25] J. Grubor, S. C. J. Lee, K.-D. Langer, T. Koonen, and J. W. Walewski, "Wireless High-Speed Data Transmission with Phosphorescent White-Light LEDs," in *33rd European Conference and Exhibition of Optical Communication - Post-Deadline Papers (published 2008)*, 2007, pp. 1–2.
- [26] H. Le Minh, D. O'Brien, G. Faulkner, L. Zeng, K. Lee, D. Jung, Y. Oh, and E. T. Won, "100-Mb/s NRZ Visible Light Communications Using a Postequalized White LED," *IEEE Photonics Technology Letters*, vol. 21, no. 15, pp. 1063–1065, 2009.
- [27] Y.-H. Chang, Y.-M. Huang, F.-J. Liou, C.-W. Chow, Y. Liu, H.-C. Kuo, C.-H. Yeh, W. H. Gunawan, T.-Y. Hung, and Y.-H. Jian, "2.805 Gbit/s high-bandwidth phosphor white light visible light communication utilizing an InGaN/GaN semipolar blue micro-LED," *Opt. Express*, vol. 30, no. 10, pp. 16938–16946, May 2022. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-30-10-16938>
- [28] H. Li, X. Chen, B. Huang, D. Tang, and H. Chen, "High Bandwidth Visible Light Communications Based on a Post-Equalization Circuit," *IEEE Photonics Technology Letters*, vol. 26, no. 2, pp. 119–122, 2014.

- [29] S.-K. Liaw, H.-H. Chou, C.-J. Wu, M.-J. Chien, and C. Teng, "500 Mb/s OOK visible light communications using RGB-based LEDs," in *2015 International Symposium on Next-Generation Electronics (ISNE)*, 2015, pp. 1–2.
- [30] H. Ma, L. Lampe, and S. Hranilovic, "Integration of indoor visible light and power line communication systems," in *2013 IEEE 17th International Symposium on Power Line Communications and Its Applications*, 2013, pp. 291–296.
- [31] J. Kahn and J. Barry, "Wireless infrared communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265–298, 1997.
- [32] L. . Melexis, "Infrared Illumination for Time-of-Flight Applications," Tech. Rep. WP35, 11 2018.
- [33] Council of European Union, "Directive 2006/25/EC of the European Parliament and of the Council of 5 April 2006 on the minimum health and safety requirements regarding the exposure of workers to risks arising from physical agents (artificial optical radiation) (19th individual Directive within the meaning of Article 16(1) of Directive 89/391/EEC)," 7 2019, <https://eur-lex.europa.eu/eli/dir/2006/25/oj/eng>.
- [34] International Electrotechnical Commission, "IEC 62471:2006 Photobiological safety of lamps and lamp systems," International Electrotechnical Commission, Standard IEC 62471:2006, 2006. [Online]. Available: <https://webstore.iec.ch/publication/7076>
- [35] European Standards, "BS EN 12464-1:2021: Light and Lighting: Lighting of work places. Indoor workplaces," European Standards, Standard BS EN 12464-1:2021, 2021. [Online]. Available: <https://www.en-standard.eu/bs-en-12464-1-2021-light-and-lighting-lighting-of-work-places-indoor-work-places/>
- [36] M. Perz, I. Vogels, D. Sekulovski, L. Wang, Y. Tu, and I. Heynderickx, "Modeling the visibility of the stroboscopic effect occurring in temporally modulated light systems," *Lighting Research & Technology*, vol. 47, no. 3, pp. 281–300, 2015. [Online]. Available: <https://doi.org/10.1177/1477153514534945>
- [37] B. Fahs, A. J. Chowdhury, Y. Zhang, J. Ghasemi, C. Hitchcock, P. Zarkesh-Ha, and M. Hella, "Blue-enhanced and bandwidth-extended photodiode in standard 0.35- μ m CMOS," 10 2016, pp. 1–3.
- [38] "Blue Enhanced Photodiodes," 2020. [Online]. Available: <https://www.osioptoelectronics.com/standard-products/silicon-photodiodes/general-purpose-photodiodes/blue-enhanced-photodiodes.aspx>
- [39] K. Verniers, L. Van der Perre, and N. Stevens, "Impact of wavelength dependency of LED and photodiode in visible light positioning," in *2017 20th International Symposium on Wireless Personal Multimedia Communications (WPMC)*, 2017, pp. 217–222.
- [40] A. Carrasco-Casado and R. Mata Calvo, *Free-space optical links for space communication networks*, 12 2020.
- [41] E. Gonendik and S. Gezici, "Fundamental Limits on RSS Based Range Estimation in Visible Light Positioning Systems," *IEEE Communications Letters*, vol. 19, no. 12, pp. 2138–2141, 2015.
- [42] S. Chhajed, Y. Xi, Y.-L. Li, T. Gessmann, and E. F. Schubert, "Influence of junction temperature on chromaticity and color-rendering properties of trichromatic white-light sources based on light-emitting diodes," *Journal of Applied Physics*, vol. 97, no. 5, p. 054506, 2005. [Online]. Available: <https://doi.org/10.1063/1.1852073>
- [43] M. Konyu, N. Ketjoy, and C. Sirisamphanwong, "Effect of dust on the solar spectrum and electricity generation of a photovoltaic module," *IET Renewable Power Generation*, vol. 14, no. 14, pp. 2759–2764, 2020. [Online]. Available: <https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/iet-rpg.2020.0456>
- [44] B. S. Yilbas, "Why environmental dust influences solar energy harvesting," *International Journal of Energy Research*, vol. 43, no. 1, pp. 4–8, 2019. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1002/er.4212>
- [45] S. Vaishak and P. V. Bhale, "Effect of dust deposition on performance characteristics of a refrigerant based photovoltaic/thermal system," *Sustainable Energy Technologies and Assessments*, vol. 36, p. 100548, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2213138819303133>
- [46] A. Y. Al-Hasan, "A new correlation for direct beam solar radiation received by photovoltaic panel with sand dust accumulated on its surface," *Solar Energy*, vol. 63, no. 5, pp. 323–333, 1998. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0038092X98000607>
- [47] S. Chaiyapinunt, B. Phueakphongsuriya, K. Mongkornsaksit, and N. Khamporn, "Performance rating of glass windows and glass windows with films in aspect of thermal comfort and heat transmission," *Energy and Buildings - ENERG BLDG*, vol. 37, pp. 725–738, 07 2005.
- [48] W. Commons. (2013) Spectrum of solar radiation at sea level (red) and out of the atmosphere (yellow). File: Solar_spectrum_en.jpg. [Online]. Available: https://commons.wikimedia.org/wiki/File:Solar_spectrum_en.svg